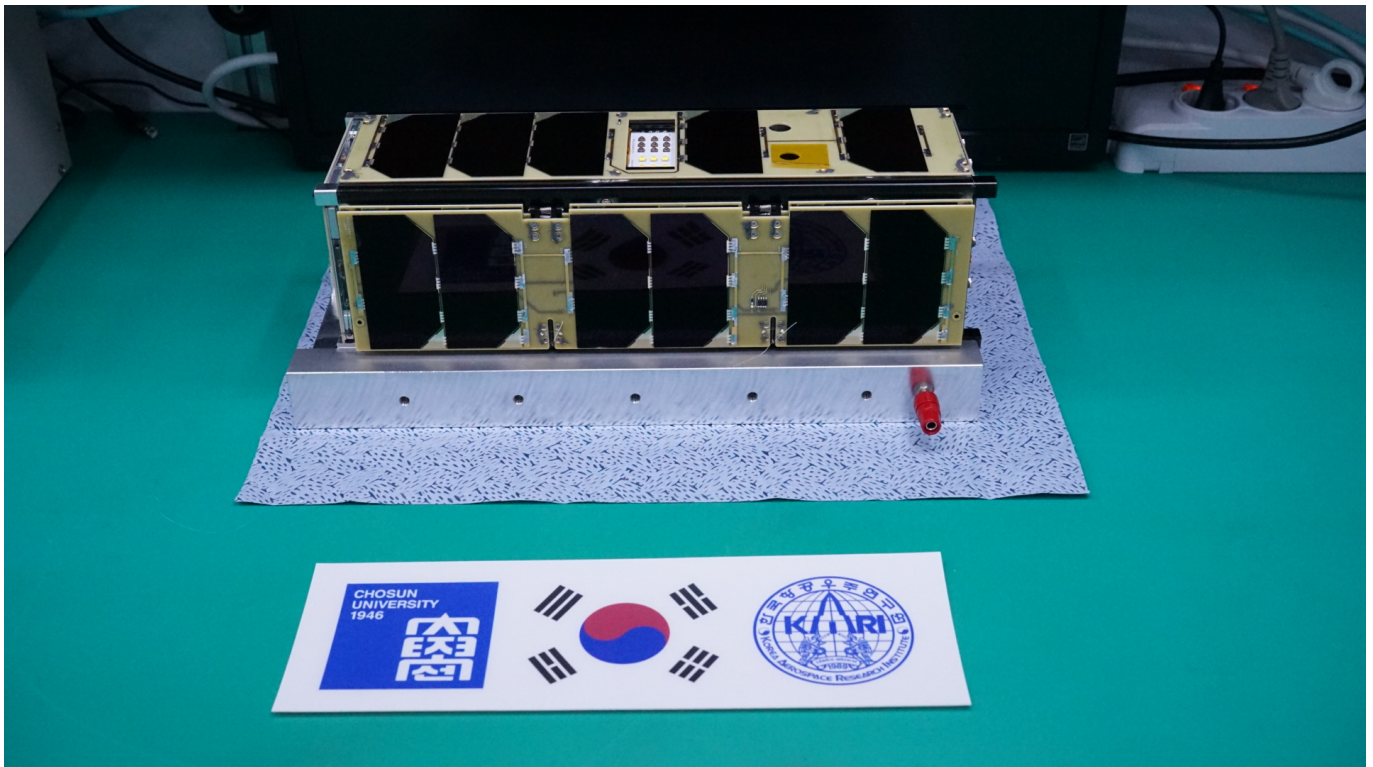


BCN Reception & Sample Validation

Cosmic Lighthouse Keeper Mission

Ten BCN modes captured from the CPSat and verified against matching waveform recordings.



CPSat project photograph supplied by the team (DSC01324.ARW embedded preview). Reference photograph; not a photograph of this RF test setup.

437.265 MHz

UHF downlink

9,600 bit/s

BPSK-G3RUH

10 BCN modes

WAV / hex verified

Pre-launch indoor over-the-air ground test; RX2; 0 dB receive gain; external attenuator removed. All 1,152,000,000 requested raw IQ samples were saved. No near-full-scale samples were observed.

01 Indoor ground-test reception

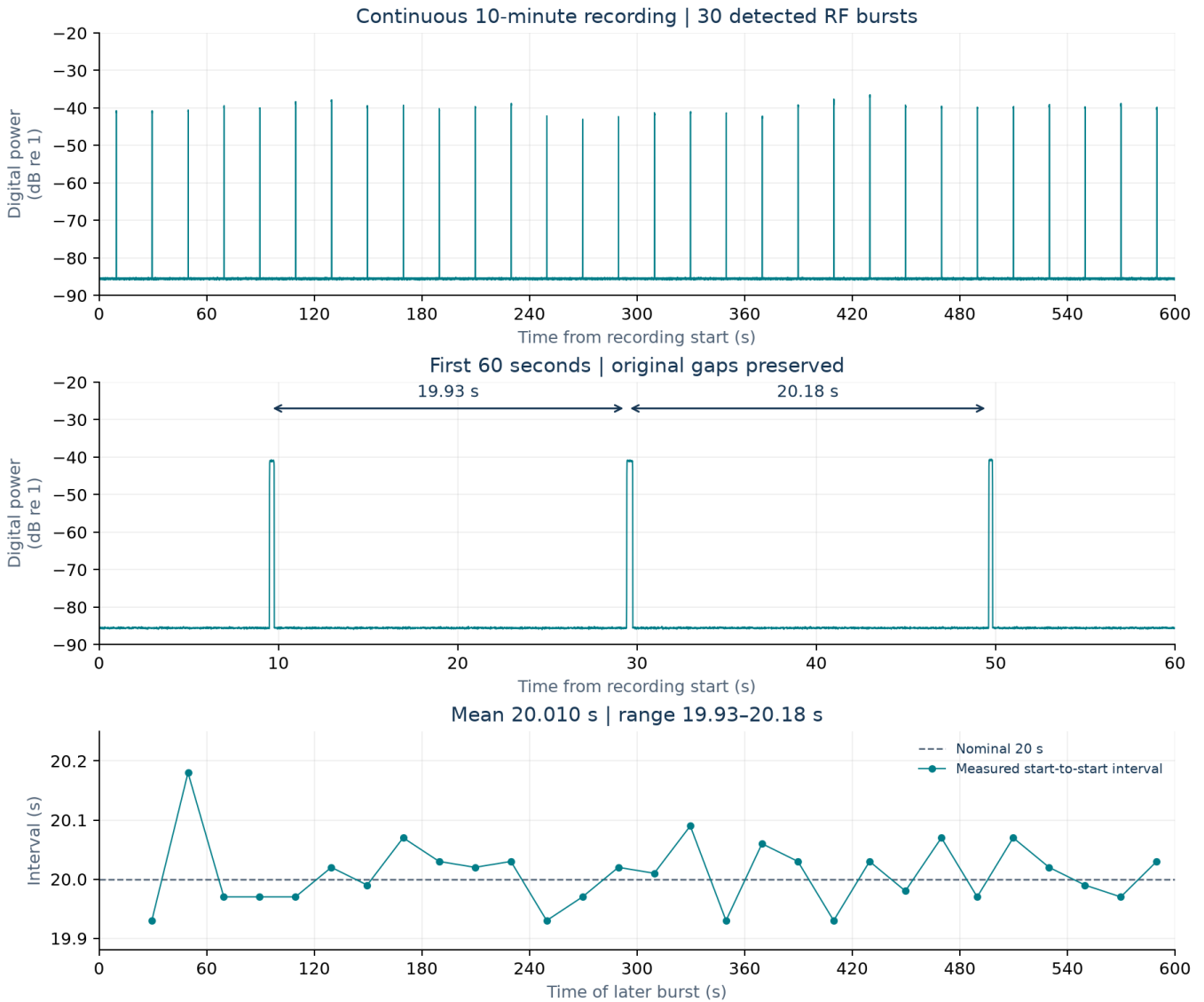


Figure 1. Continuous 600-second reception, including the original intervals without a beacon. Power is averaged over consecutive 10 ms windows of the original 192 ksamples/s complex channel; no amplitude normalization is applied. A threshold 20 dB above median recording power detects 30 RF bursts and 29 start-to-start intervals. Thresholds of 15 and 25 dB also give 30 bursts. Times lie on a 10 ms detection grid. These are energy detections, not counts of decoded packets or exact transmitter scheduling timestamps.

Measured mean interval: **20.010 s**; range: **19.93–20.18 s**. Each narrow peak in the full view is a short RF burst. The first-minute view preserves the long gaps between bursts. Source values are provided in `data/continuous_power.csv` and `data/burst_timing.csv`.

02 BPSK recovery and decoded BCN

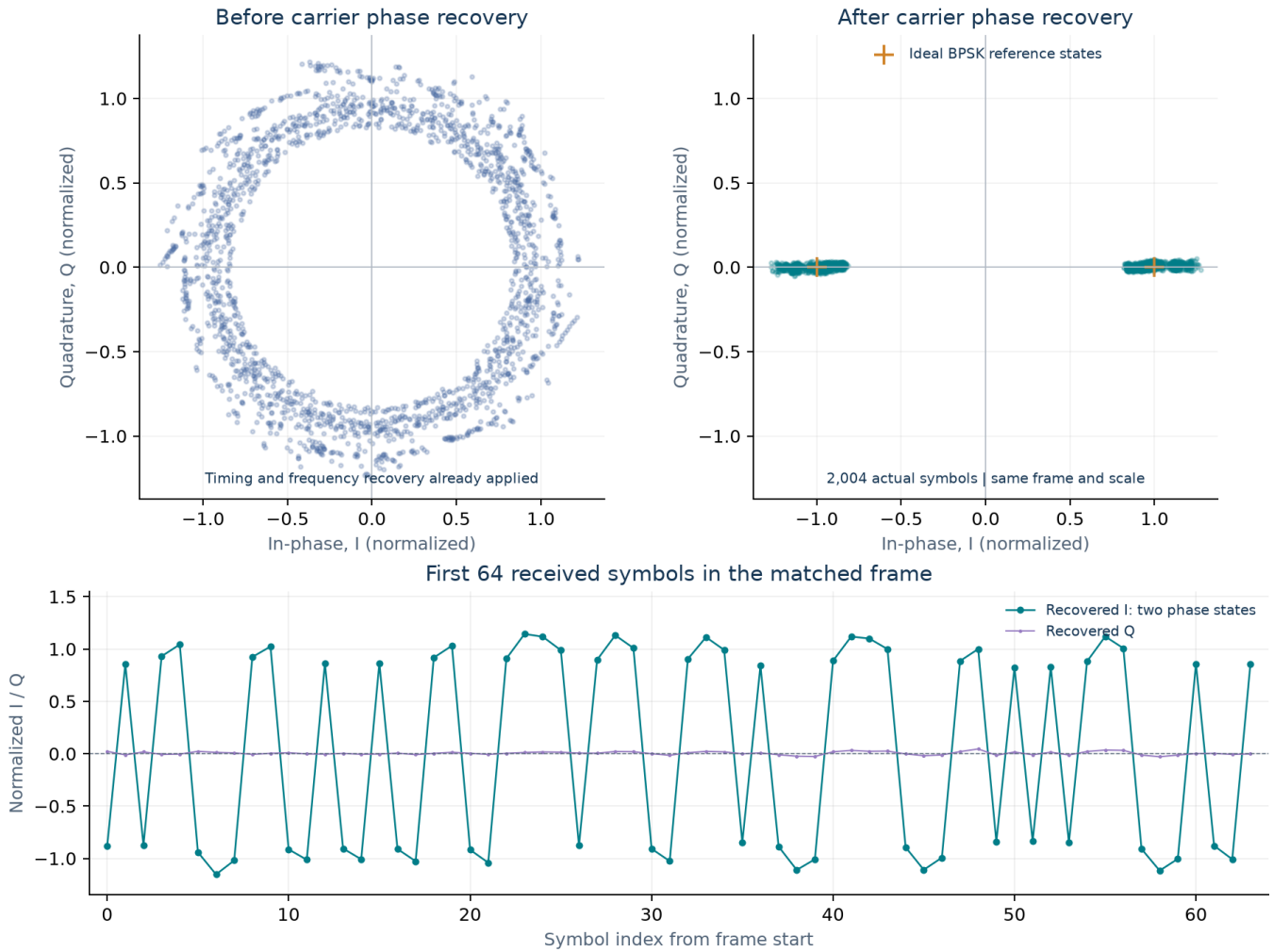


Figure 2. Actual mode 02 symbols before and after the order-2 Costas carrier phase recovery loop. Each point is one symbol. Both views use the same 2,004 symbols of the complete matched, HDLC-stuffed frame including FCS, with one shared RMS amplitude scale. No additional phase rotation or point removal is applied. Silence and preamble are outside this frame selection. The source WAV already includes its documented +870 Hz correction; timing and frequency recovery precede the left-hand trace.

The two clusters after recovery show that the receiver separates the two BPSK phase states. The lower trace makes their alternating signs visible. Those signs are channel symbols; NRZI decoding and G3RUH descrambling are still required before interpreting AX.25 bits. The resulting frame passes FCS and matches the published hex byte for byte. This is one successfully decoded example, not a BER, calibrated EVM, reception-rate or sensitivity measurement.

Check	Observed result
AX.25 FCS	PASS
Destination / source	CPSAT1-0 / CPSAT1-0
AX.25 data / CCSDS BCN	248 bytes without FCS / 232 bytes
BCN identity	MID 0x0903; mode 02 / COMMON
Published sample comparison	Exact AX.25 hex match

03 Single-beacon detail

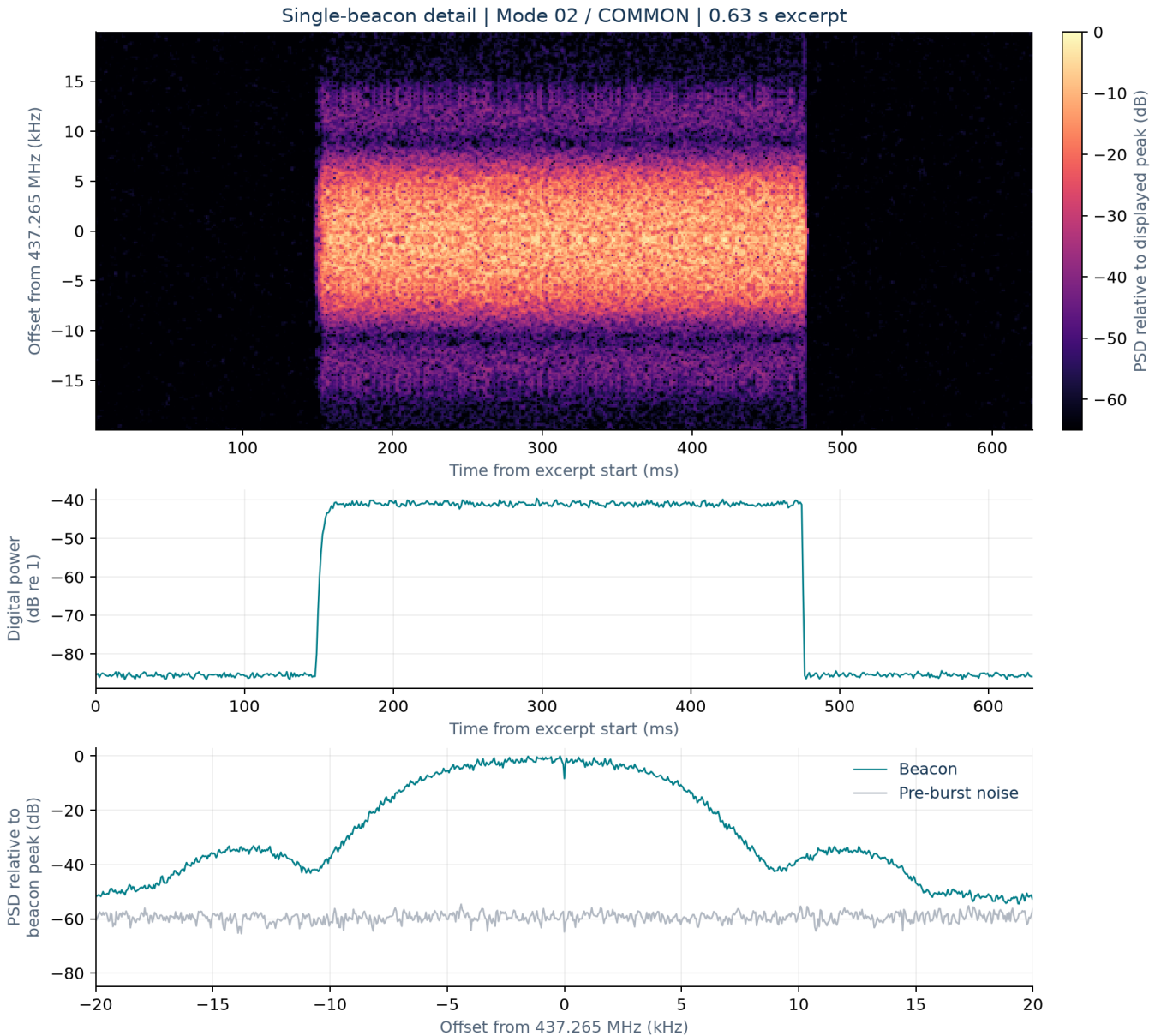


Figure 3. One mode 02 beacon in a 0.63-second excerpt; this view does not show the 20-second repetition interval. The plotted IQ is reconstructed from the supplied PCM WAV by reversing its documented amplitude scale and +870 Hz frequency correction. PCM quantization remains. Digital levels are not calibrated dBm. Waterfall: Hann window, 1,024 samples, 75% overlap; spectrum: Welch, 4,096 samples.

Nominal beacon interval: 20 seconds. The figure shows one selected burst, with time relative to its excerpt start. It does not represent a continuous pass or an orbital reception.

04 BCN coverage and validation

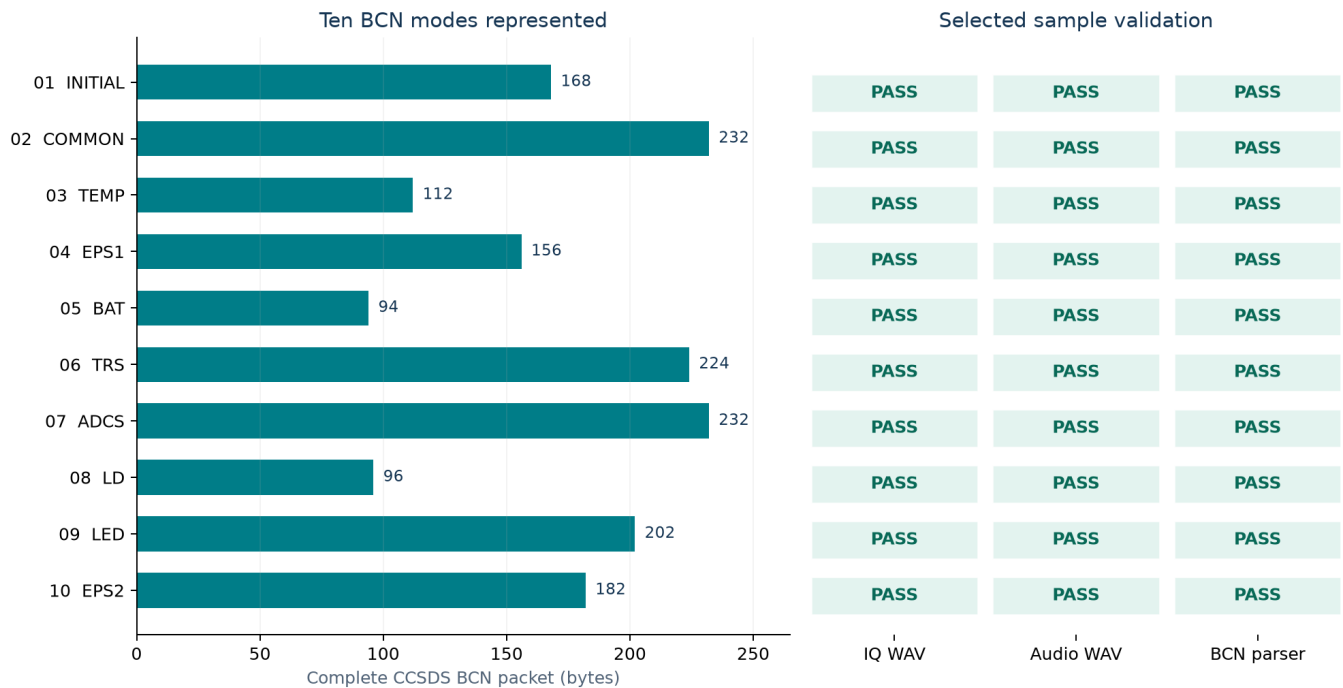


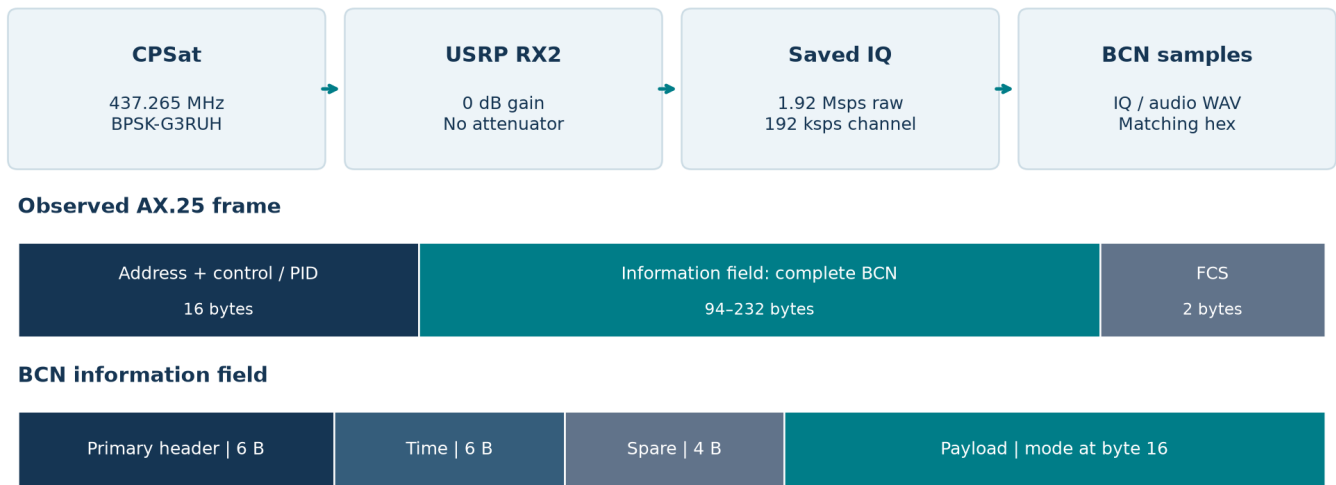
Figure 4. One selected example for each BCN mode. Both WAV formats independently reproduce the same CRC-verified AX.25 hex as the corresponding complex excerpt. All ten packets also pass the existing BCN parser. This is sample validation, not a packet reception rate.

The first offline analysis recovered 20 CRC-valid BCN frames across all ten modes from 21 burst excerpts in approximately the first 410 seconds. The package selects one verified example per mode. Nine selected examples include a +870 Hz digital frequency correction; mode 04 retains the original offset. Per-sample amplitude scales and hashes are in manifest.json. These are Ground-test samples, not in-orbit observations.

Mode	BCN packet	CCSDS bytes	AX.25 bytes*
01	BCN_INITIAL_PKT	168	184
02	BCN_COMMON_PKT	232	248
03	BCN_TEMP_PKT	112	128
04	BCN_EPS1_PKT	156	172
05	BCN_BAT_PKT	94	110
06	BCN_TRS_PKT	224	240
07	BCN_ADCS_PKT	232	248
08	BCN_LD_PKT	96	112
09	BCN_LED_PKT	202	218
10	BCN_EPS2_PKT	182	198

* AX.25 bytes in the supplied hex, excluding FCS and flags.

05 Framing and sample files



Schematic, not to scale. FCS is checked and removed from supplied AX.25 hex; flags are omitted.

Figure 5. Receive-only recording path and observed packet structure. The supplied samples contain complete BCN packets directly in the AX.25 information field; no reassembly was required.

iq_192k.wav — 192 kHz stereo PCM16; channel 1 = I, channel 2 = Q.

audio_48k.wav — 48 kHz mono PCM16; real BPSK passband near +12 kHz.

ax25_no_fcs.hex — one CRC-verified frame, with FCS and flags removed.

bcn_ccsds.hex — complete information field, including the original 16-byte header.

Addressing: destination and source CPSAT1-0; control 0x03; PID 0xF0. BCN MID 0x0903; mode at zero-based byte 16. Spacecraft time is retained as raw fields; the epoch is unconfirmed.

Initial live decoding did not recover valid BCN. Offline processing removed the extra fast AGC and reset synchronization for each burst. The source recording is preserved. This report validates the selected files; continuous live-decoder performance has not been established.

Reproduction details and per-file checksums: README.md and manifest.json. SatNOGS integration remains subject to maintainer review.

The following ten pages show each actual beacon as an annotated hex dump. Offsets refer to supplied AX.25 bytes without FCS. Protocol references: AX.25 v2.2 and CCSDS Space Packet Protocol.

Hex guide | Mode 01 / BCN_INITIAL_PKT

All offsets: hexadecimal, zero-based, inclusive. Entire AX.25 sample: 184 bytes, excluding FCS. The AX.25 header ends at 0x000F; the information field / CCSDS packet occupies 0x0010–0x00B7.

Field	Frame offset	CCSDS offset	Actual hex
Destination address	0000–0006	—	86 A0 A6 82 A8 62 E0
Source address	0007–000D	—	86 A0 A6 82 A8 62 61
Control (UI)	000E	—	03
PID	000F	—	F0
CCSDS packet ID / MID	0010–0011	0000–0001	09 03
CCSDS sequence control	0012–0013	0002–0003	C0 0B
CCSDS packet length	0014–0015	0004–0005	00 A1
Seconds (raw)	0016–0019	0006–0009	57 F7 32 8A
Subseconds (raw)	001A–001B	000A–000B	84 BD
Spare / alignment	001C–001F	000C–000F	00 00 00 00
BCN mode (payload byte 0)	0020	0010	01
Remaining BCN payload	0021–00B7	0011–00A7	See complete dump below

CCSDS primary header: **09 03 C0 0B 00 A1**. MID 0x0903 / APID 0x103; sequence flags 11 (unsegmented), count 11. Length 0x00A1 (161) + 7 = 168 bytes. Time values are raw; no epoch is assumed.

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	86	A0	A6	82	A8	62	E0	86	A0	A6	82	A8	62	61	03	F0
0010	09	03	C0	0B	00	A1	57	F7	32	8A	84	BD	00	00	00	00
0020	01	01	01	00	00	FE	00	00	FF	0C	CF	00	00	00	00	80
0030	00	07	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00
0040	00	00	00	00	0A	01	02	03	04	05	06	07	08	09	0A	00
0050	00	00	00	00	00	00	00	02	6D	03	21	00	FF	FF	FF	FF
0060	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF
0070	FF	FF	FF	FF	01	00	00	00	00	00	00	00	00	00	00	00
0080	00	00	80	05	00	00	00	00	00	00	FF	7F	00	00	00	00
0090	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00A0	3B	0B	00	00	FF	FF	FF	00	00	01	00	00	00	01	00	1F
00B0	00	00	00	00	00	00	FF	FF								

Colors: AX.25 header (blue), primary header (lavender), time (light blue), spare/alignment (gray), mode (orange), remaining payload (green). The mode byte belongs to the payload. Header and information together form AX.25; no FCS or flags are included in this dump.

Hex guide | Mode 02 / BCN_COMMON_PKT

All offsets: hexadecimal, zero-based, inclusive. Entire AX.25 sample: 248 bytes, excluding FCS. The AX.25 header ends at 0x000F; the information field / CCSDS packet occupies 0x0010–0x00F7.

Field	Frame offset	CCSDS offset	Actual hex
Destination address	0000–0006	—	86 A0 A6 82 A8 62 E0
Source address	0007–000D	—	86 A0 A6 82 A8 62 61
Control (UI)	000E	—	03
PID	000F	—	F0
CCSDS packet ID / MID	0010–0011	0000–0001	09 03
CCSDS sequence control	0012–0013	0002–0003	C0 0C
CCSDS packet length	0014–0015	0004–0005	00 E1
Seconds (raw)	0016–0019	0006–0009	57 F7 32 9E
Subseconds (raw)	001A–001B	000A–000B	82 B7
Spare / alignment	001C–001F	000C–000F	00 00 00 00
BCN mode (payload byte 0)	0020	0010	02
Remaining BCN payload	0021–00F7	0011–00E7	See complete dump below

CCSDS primary header: **09 03 C0 0C 00 E1**. MID 0x0903 / APID 0x103; sequence flags 11 (unsegmented), count 12. Length 0x00E1 (225) + 7 = 232 bytes. Time values are raw; no epoch is assumed.

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	86	A0	A6	82	A8	62	E0	86	A0	A6	82	A8	62	61	03	F0
0010	09	03	C0	0C	00	E1	57	F7	32	9E	82	B7	00	00	00	00
0020	02	01	00	00	00	00	00	00	01	E2	00	00	12	00	00	00
0030	00	00	00	00	00	00	00	00	6D	03	22	00	5A	03	05	00
0040	03	03	07	00	00	00	00	00	00	00	32	00	6E	00	97	FF
0050	C6	03	17	03	36	02	01	00	00	00	66	00	00	00	00	00
0060	00	00	00	00	00	00	00	00	8C	35	02	00	8D	31	F7	57
0070	21	01	00	00	21	01	00	00	00	00	00	00	00	00	00	00
0080	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0090	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00A0	02	02	00	01	CE	1E	00	47	00	00	62	00	1E	00	00	00
00B0	04	03	0B	00	AA	00	00	00	04	03	0B	00	A3	00	00	00
00C0	04	03	0B	00	B8	00	00	00	04	03	0B	00	B1	00	00	00
00D0	00	00	00	00	00	00	00	00	00	00	13	01	00	00	00	00
00E0	00	01	01	00	00	00	00	00	00	00	00	00	04	00	01	00
00F0	00	00	00	00	09	00	00	00								

Colors: AX.25 header (blue), primary header (lavender), time (light blue), spare/alignment (gray), mode (orange), remaining payload (green). The mode byte belongs to the payload. Header and information together form AX.25; no FCS or flags are included in this dump.

Hex guide | Mode 03 / BCN_TEMP_PKT

All offsets: hexadecimal, zero-based, inclusive. Entire AX.25 sample: 128 bytes, excluding FCS. The AX.25 header ends at 0x000F; the information field / CCSDS packet occupies 0x0010–0x007F.

Field	Frame offset	CCSDS offset	Actual hex
Destination address	0000–0006	—	86 A0 A6 82 A8 62 E0
Source address	0007–000D	—	86 A0 A6 82 A8 62 61
Control (UI)	000E	—	03
PID	000F	—	F0
CCSDS packet ID / MID	0010–0011	0000–0001	09 03
CCSDS sequence control	0012–0013	0002–0003	C0 0D
CCSDS packet length	0014–0015	0004–0005	00 69
Seconds (raw)	0016–0019	0006–0009	57 F7 32 B2
Subseconds (raw)	001A–001B	000A–000B	84 B9
Spare / alignment	001C–001F	000C–000F	00 00 00 00
BCN mode (payload byte 0)	0020	0010	03
Remaining BCN payload	0021–007F	0011–006F	See complete dump below

CCSDS primary header: **09 03 C0 0D 00 69**. MID 0x0903 / APID 0x103; sequence flags 11 (unsegmented), count 13. Length 0x0069 (105) + 7 = 112 bytes. Time values are raw; no epoch is assumed.

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	86	A0	A6	82	A8	62	E0	86	A0	A6	82	A8	62	61	03	F0
0010	09	03	C0	0D	00	69	57	F7	32	B2	84	B9	00	00	00	00
0020	03	80	00	00	00	00	1C	03	94	02	94	02	93	02	92	02
0030	F3	03	F7	03	F5	03	F9	03	F7	03	F5	03	9D	02	9E	02
0040	9E	02	9B	02	9D	02	9E	02	9E	02	9E	02	9C	02	9E	02
0050	9D	02	9E	02	2B	03	23	03	FF	FF	C2	08	B7	08	00	01
0060	00	00	00	00	00	00	00	00	1D	03	1C	03	94	02	93	02
0070	2C	03	29	03	CA	08	01	00	02	00	00	00	F0	06	84	00

Colors: AX.25 header (blue), primary header (lavender), time (light blue), spare/alignment (gray), mode (orange), remaining payload (green). The mode byte belongs to the payload. Header and information together form AX.25; no FCS or flags are included in this dump.

Hex guide | Mode 04 / BCN_EPS1_PKT

All offsets: hexadecimal, zero-based, inclusive. Entire AX.25 sample: 172 bytes, excluding FCS. The AX.25 header ends at 0x000F; the information field / CCSDS packet occupies 0x0010–0x00AB.

Field	Frame offset	CCSDS offset	Actual hex
Destination address	0000–0006	—	86 A0 A6 82 A8 62 E0
Source address	0007–000D	—	86 A0 A6 82 A8 62 61
Control (UI)	000E	—	03
PID	000F	—	F0
CCSDS packet ID / MID	0010–0011	0000–0001	09 03
CCSDS sequence control	0012–0013	0002–0003	C0 0E
CCSDS packet length	0014–0015	0004–0005	00 95
Seconds (raw)	0016–0019	0006–0009	57 F7 32 C6
Subseconds (raw)	001A–001B	000A–000B	86 C8
Spare / alignment	001C–001F	000C–000F	00 00 00 00
BCN mode (payload byte 0)	0020	0010	04
Remaining BCN payload	0021–00AB	0011–009B	See complete dump below

CCSDS primary header: **09 03 C0 0E 00 95**. MID 0x0903 / APID 0x103; sequence flags 11 (unsegmented), count 14. Length 0x0095 (149) + 7 = 156 bytes. Time values are raw; no epoch is assumed.

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	86	A0	A6	82	A8	62	E0	86	A0	A6	82	A8	62	61	03	F0
0010	09	03	C0	0E	00	95	57	F7	32	C6	86	C8	00	00	00	00
0020	04	00	00	E2	0F	00	00	00	00	00	00	00	C0	02	C0	02
0030	80	00	3D	00	71	03	14	00	10	00	02	00	7C	03	06	00
0040	6B	03	05	00	5C	03	5B	00	06	03	02	00	02	00	02	00
0050	02	00	02	00	02	00	5D	03	05	03	0F	00	06	03	02	00
0060	02	00	02	00	02	00	02	00	02	00	03	00	06	00	02	00
0070	03	00	2C	03	24	03	54	00	17	00	DC	07	00	00	02	00
0080	FF	00	02	00	05	00	5B	00	06	00	6B	03	2C	03	2C	03
0090	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00A0	82	14	00	00	ED	D1	00	00	48	00	00	00				

Colors: AX.25 header (blue), primary header (lavender), time (light blue), spare/alignment (gray), mode (orange), remaining payload (green). The mode byte belongs to the payload. Header and information together form AX.25; no FCS or flags are included in this dump.

Hex guide | Mode 05 / BCN_BAT_PKT

All offsets: hexadecimal, zero-based, inclusive. Entire AX.25 sample: 110 bytes, excluding FCS. The AX.25 header ends at 0x000F; the information field / CCSDS packet occupies 0x0010–0x006D.

Field	Frame offset	CCSDS offset	Actual hex
Destination address	0000–0006	—	86 A0 A6 82 A8 62 E0
Source address	0007–000D	—	86 A0 A6 82 A8 62 61
Control (UI)	000E	—	03
PID	000F	—	F0
CCSDS packet ID / MID	0010–0011	0000–0001	09 03
CCSDS sequence control	0012–0013	0002–0003	C0 0F
CCSDS packet length	0014–0015	0004–0005	00 57
Seconds (raw)	0016–0019	0006–0009	57 F7 32 DA
Subseconds (raw)	001A–001B	000A–000B	84 B9
Spare / alignment	001C–001F	000C–000F	00 00 00 00
BCN mode (payload byte 0)	0020	0010	05
Remaining BCN payload	0021–006D	0011–005D	See complete dump below

CCSDS primary header: **09 03 C0 0F 00 57**. MID 0x0903 / APID 0x103; sequence flags 11 (unsegmented), count 15. Length 0x0057 (87) + 7 = 94 bytes. Time values are raw; no epoch is assumed.

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	86	A0	A6	82	A8	62	E0	86	A0	A6	82	A8	62	61	03	F0
0010	09	03	C0	0F	00	57	57	F7	32	DA	84	B9	00	00	00	00
0020	05	00	00	00	00	11	00	09	01	80	00	00	36	08	B5	35
0030	6E	03	24	00	08	00	1D	03	05	00	59	03	07	00	03	03
0040	95	02	94	02	94	02	92	02	02	00	02	00	02	00	02	00
0050	6E	03	57	03	2A	00	1D	03	1C	03	95	02	95	02	95	02
0060	93	02	04	00	00	00	00	01	00	00	00	00	00	00		

Colors: AX.25 header (blue), primary header (lavender), time (light blue), spare/alignment (gray), mode (orange), remaining payload (green). The mode byte belongs to the payload. Header and information together form AX.25; no FCS or flags are included in this dump.

Hex guide | Mode 06 / BCN_TRS_PKT

All offsets: hexadecimal, zero-based, inclusive. Entire AX.25 sample: 240 bytes, excluding FCS. The AX.25 header ends at 0x000F; the information field / CCSDS packet occupies 0x0010–0x00EF.

Field	Frame offset	CCSDS offset	Actual hex
Destination address	0000–0006	—	86 A0 A6 82 A8 62 E0
Source address	0007–000D	—	86 A0 A6 82 A8 62 61
Control (UI)	000E	—	03
PID	000F	—	F0
CCSDS packet ID / MID	0010–0011	0000–0001	09 03
CCSDS sequence control	0012–0013	0002–0003	C0 10
CCSDS packet length	0014–0015	0004–0005	00 D9
Seconds (raw)	0016–0019	0006–0009	57 F7 32 EE
Subseconds (raw)	001A–001B	000A–000B	84 B3
Spare / alignment	001C–001F	000C–000F	00 00 00 00
BCN mode (payload byte 0)	0020	0010	06
Remaining BCN payload	0021–00EF	0011–00DF	See complete dump below

CCSDS primary header: **09 03 C0 10 00 D9**. MID 0x0903 / APID 0x103; sequence flags 11 (unsegmented), count 16. Length 0x00D9 (217) + 7 = 224 bytes. Time values are raw; no epoch is assumed.

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	86	A0	A6	82	A8	62	E0	86	A0	A6	82	A8	62	61	03	F0
0010	09	03	C0	10	00	D9	57	F7	32	EE	84	B3	00	00	00	00
0020	06	00	00	00	AA	05	77	06	47	00	28	00	4E	06	35	01
0030	43	00	4A	02	01	00	C4	08	B7	08	00	00	00	00	0C	0C
0040	00	00	00	00	6E	01	00	00	6E	01	00	00	6E	01	0C	01
0050	AB	02	00	00	00	00	00	00	00	00	00	00	D6	08	41	06
0060	FA	05	2B	0E	C1	03	CA	08	6E	01	0C	01	00	00	00	00
0070	AB	39	02	00	11	AC	06	00	11	AC	06	00	4E	06	35	00
0080	11	AC	06	00	4E	06	35	00	00	00	00	00	00	00	00	00
0090	B8	08	00	00	00	00	D9	03	64	0D	00	00	00	00	00	00
00A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00B0	00	00	00	00	8C	31	F7	57	8C	31	F7	57	2B	0E	4E	0C
00C0	C1	03	CA	08	BE	08	C4	08	B7	08	C5	08	B8	08	D6	08
00D0	00	00	FA	05	00	00	00	00	12	01	00	00	00	00	00	00
00E0	C4	09	00	00	00	00	00	00	00	00	00	27	01	03	00	00

Colors: AX.25 header (blue), primary header (lavender), time (light blue), spare/alignment (gray), mode (orange), remaining payload (green). The mode byte belongs to the payload. Header and information together form AX.25; no FCS or flags are included in this dump.

Hex guide | Mode 07 / BCN_ADCS_PKT

All offsets: hexadecimal, zero-based, inclusive. Entire AX.25 sample: 248 bytes, excluding FCS. The AX.25 header ends at 0x000F; the information field / CCSDS packet occupies 0x0010–0x00F7.

Field	Frame offset	CCSDS offset	Actual hex
Destination address	0000–0006	—	86 A0 A6 82 A8 62 E0
Source address	0007–000D	—	86 A0 A6 82 A8 62 61
Control (UI)	000E	—	03
PID	000F	—	F0
CCSDS packet ID / MID	0010–0011	0000–0001	09 03
CCSDS sequence control	0012–0013	0002–0003	C0 1B
CCSDS packet length	0014–0015	0004–0005	00 E1
Seconds (raw)	0016–0019	0006–0009	57 F7 33 CA
Subseconds (raw)	001A–001B	000A–000B	85 1A
Spare / alignment	001C–001F	000C–000F	00 00 00 00
BCN mode (payload byte 0)	0020	0010	07
Remaining BCN payload	0021–00F7	0011–00E7	See complete dump below

CCSDS primary header: **09 03 C0 1B 00 E1**. MID 0x0903 / APID 0x103; sequence flags 11 (unsegmented), count 27. Length 0x00E1 (225) + 7 = 232 bytes. Time values are raw; no epoch is assumed.

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	86	A0	A6	82	A8	62	E0	86	A0	A6	82	A8	62	61	03	F0
0010	09	03	C0	1B	00	E1	57	F7	33	CA	85	1A	00	00	00	00
0020	07	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0030	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0040	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0050	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0060	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0070	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0080	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0090	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00D0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00F0	00	00	00	00	00	00	00	00								

Colors: AX.25 header (blue), primary header (lavender), time (light blue), spare/alignment (gray), mode (orange), remaining payload (green). The mode byte belongs to the payload. Header and information together form AX.25; no FCS or flags are included in this dump.

Hex guide | Mode 08 / BCN_LD_PKT

All offsets: hexadecimal, zero-based, inclusive. Entire AX.25 sample: 112 bytes, excluding FCS. The AX.25 header ends at 0x000F; the information field / CCSDS packet occupies 0x0010–0x006F.

Field	Frame offset	CCSDS offset	Actual hex
Destination address	0000–0006	—	86 A0 A6 82 A8 62 E0
Source address	0007–000D	—	86 A0 A6 82 A8 62 61
Control (UI)	000E	—	03
PID	000F	—	F0
CCSDS packet ID / MID	0010–0011	0000–0001	09 03
CCSDS sequence control	0012–0013	0002–0003	C0 1C
CCSDS packet length	0014–0015	0004–0005	00 59
Seconds (raw)	0016–0019	0006–0009	57 F7 33 DE
Subseconds (raw)	001A–001B	000A–000B	83 3E
Spare / alignment	001C–001F	000C–000F	00 00 00 00
BCN mode (payload byte 0)	0020	0010	08
Remaining BCN payload	0021–006F	0011–005F	See complete dump below

CCSDS primary header: **09 03 C0 1C 00 59**. MID 0x0903 / APID 0x103; sequence flags 11 (unsegmented), count 28. Length 0x0059 (89) + 7 = 96 bytes. Time values are raw; no epoch is assumed.

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	86	A0	A6	82	A8	62	E0	86	A0	A6	82	A8	62	61	03	F0
0010	09	03	C0	1C	00	59	57	F7	33	DE	83	3E	00	00	00	00
0020	08	67	01	00	00	00	00	00	00	00	00	00	00	00	00	00
0030	AA	DD	25	11	00	00	00	00	00	00	00	00	00	00	00	00
0040	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0050	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0060	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Colors: AX.25 header (blue), primary header (lavender), time (light blue), spare/alignment (gray), mode (orange), remaining payload (green). The mode byte belongs to the payload. Header and information together form AX.25; no FCS or flags are included in this dump.

Hex guide | Mode 09 / BCN_LED_PKT

All offsets: hexadecimal, zero-based, inclusive. Entire AX.25 sample: 218 bytes, excluding FCS. The AX.25 header ends at 0x000F; the information field / CCSDS packet occupies 0x0010–0x00D9.

Field	Frame offset	CCSDS offset	Actual hex
Destination address	0000–0006	—	86 A0 A6 82 A8 62 E0
Source address	0007–000D	—	86 A0 A6 82 A8 62 61
Control (UI)	000E	—	03
PID	000F	—	F0
CCSDS packet ID / MID	0010–0011	0000–0001	09 03
CCSDS sequence control	0012–0013	0002–0003	C0 1D
CCSDS packet length	0014–0015	0004–0005	00 C3
Seconds (raw)	0016–0019	0006–0009	57 F7 33 F2
Subseconds (raw)	001A–001B	000A–000B	84 BE
Spare / alignment	001C–001F	000C–000F	00 00 00 00
BCN mode (payload byte 0)	0020	0010	09
Remaining BCN payload	0021–00D9	0011–00C9	See complete dump below

CCSDS primary header: **09 03 C0 1D 00 C3**. MID 0x0903 / APID 0x103; sequence flags 11 (unsegmented), count 29. Length 0x00C3 (195) + 7 = 202 bytes. Time values are raw; no epoch is assumed.

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	86	A0	A6	82	A8	62	E0	86	A0	A6	82	A8	62	61	03	F0
0010	09	03	C0	1D	00	C3	57	F7	33	F2	84	BE	00	00	00	00
0020	09	00	00	00	00	00	00	00	FF	FF	00	00	00	00	00	00
0030	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0040	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0050	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0060	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0070	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
0080	00	FF	00	00	FF	FF	00	00	00	00	00	00	00	00	FF	00
0090	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00A0	00	00	00	00	00	00	00	00	00	00	00	00	FF	FF	FF	FF
00B0	FF	F0	00	00	00	00	00	00	00	00	00	00	00	00	00	0F
00C0	FF	F0	00	0F	FF	F0	00	00	00	00	00	00	00	00	00	00
00D0	00	00	00	00	00	00	04	01	00	00						

Colors: AX.25 header (blue), primary header (lavender), time (light blue), spare/alignment (gray), mode (orange), remaining payload (green). The mode byte belongs to the payload. Header and information together form AX.25; no FCS or flags are included in this dump.

Hex guide | Mode 10 / BCN_EPS2_PKT

All offsets: hexadecimal, zero-based, inclusive. Entire AX.25 sample: 198 bytes, excluding FCS. The AX.25 header ends at 0x000F; the information field / CCSDS packet occupies 0x0010–0x00C5.

Field	Frame offset	CCSDS offset	Actual hex
Destination address	0000–0006	—	86 A0 A6 82 A8 62 E0
Source address	0007–000D	—	86 A0 A6 82 A8 62 61
Control (UI)	000E	—	03
PID	000F	—	F0
CCSDS packet ID / MID	0010–0011	0000–0001	09 03
CCSDS sequence control	0012–0013	0002–0003	C0 14
CCSDS packet length	0014–0015	0004–0005	00 AF
Seconds (raw)	0016–0019	0006–0009	57 F7 33 3E
Subseconds (raw)	001A–001B	000A–000B	84 BE
Spare / alignment	001C–001F	000C–000F	00 00 00 00
BCN mode (payload byte 0)	0020	0010	0A
Remaining BCN payload	0021–00C5	0011–00B5	See complete dump below

CCSDS primary header: **09 03 C0 14 00 AF**. MID 0x0903 / APID 0x103; sequence flags 11 (unsegmented), count 20. Length 0x00AF (175) + 7 = 182 bytes. Time values are raw; no epoch is assumed.

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0000	86	A0	A6	82	A8	62	E0	86	A0	A6	82	A8	62	61	03	F0
0010	09	03	C0	14	00	AF	57	F7	33	3E	84	BE	00	00	00	00
0020	0A	00	10	00	02	00	02	00	02	00	02	00	F3	03	F8	03
0030	02	00	02	00	02	00	02	00	02	00	F4	03	F9	03	02	00
0040	02	00	02	00	02	00	02	00	F7	03	F4	03	02	00	02	00
0050	FE	00	02	00	02	00	9E	02	9F	02	02	00	02	00	FE	00
0060	02	00	02	00	9E	02	9C	02	02	00	02	00	FF	00	02	00
0070	02	00	9D	02	A0	02	02	00	02	00	EB	00	02	00	02	00
0080	9D	02	9E	02	02	00	02	00	FE	00	02	00	02	00	9C	02
0090	9E	02	02	00	02	00	35	01	7A	01	8B	01	9D	02	9E	02
00A0	02	00	02	00	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00
00B0	00	00	00	00	0E	0D	00	0C	00	00	00	00	00	00	00	00
00C0	00	00	10	00	00	00										

Colors: AX.25 header (blue), primary header (lavender), time (light blue), spare/alignment (gray), mode (orange), remaining payload (green). The mode byte belongs to the payload. Header and information together form AX.25; no FCS or flags are included in this dump.